

## EE 520: Microelectronics Devices and Modelling

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| Course Code   | EE 520                                                                                            |
| Course Name   | Microelectronics Devices and Modelling                                                            |
| L-T-P-C       | 3-0-0-3                                                                                           |
| Prerequisites | Applied Electronics (IC 161 & IC 161 P) and Device Electronics for Integrated (EE311 and EE312 P) |
| Intended for  | M.Tech EE (VLSI) / UG                                                                             |
| Distribution  | Core for M.Tech EE (VLSI) and Elective for UG                                                     |
| Approval      | 10th Senate (Jun 2016)                                                                            |

### Course Contents

- **SEMICONDUCTOR ELECTRONICS:**
  - Physics of Semiconductor Materials, Band Model of Solids Thermal-Equilibrium Statistics, Carriers in Semiconductors, Drift Velocity, Mobility and Scattering, Drift & Diffusion Current, Device: Hall-Effect. [6 Lectures]
- **METAL-SEMICONDUCTOR CONTACTS and P-N JUNCTIONS:**
  - Metal-Semiconductor junctions, Current-Voltage Characteristics, Surface Effects. The pn junction, Step Junction, Linearly Graded Junction, Heterojunctions, Reverse-Biased p-n junctions and break down mechanism. Generation and Recombination. [8 Lectures]
- **FIELD-EFFECT TRANSISTORS (MOSFETs): PHYSICAL EFFECTS AND MODELS:**
  - MOS Capacitor, Oxide and Interface Charge: Origin and Experimental Determination Charge- Coupled Devices, non-volatile memory.
  - Basic MOSFET behaviour, MOSFET scaling and short channel model. Devices: Complementary MOSFETs (CMOS), electric fields and velocity-saturation, basic leakage currents, channel length modulation, body bias effect, threshold adjustment, sub-threshold conduction. [9 Lectures]
- **Device Modeling:**
  - Limitation of long channel analysis, short-channel effects: velocity saturation, device degradation, channel length modulation, body bias effect, threshold adjustment, mobility degradation, hot carrier effects, MOSFET scaling goals, gate coupling, velocity overshoot, high field effects in scaled MOSFETs, substrate current and effects in scaled MOSFETs.
  - Moore law, Technology nodes and ITRS, Physical & Technological Challenges to scaling, Nonconventional MOSFET (FDSOI, SOI, Multi-gate MOSFETs). [10 Lectures]
- **Numerical Simulation:**
  - Numerical simulation, basic concepts of simulations, grids, device simulation and challenges. Importance of semiconductor device simulators - Key elements of physical device simulation, historical development of the physical device modeling.
  - Introduction to the TCAD Simulation Tool, Examples of TCAD Simulations MOSFETs and SOI. [8 Lectures]

### Text Books

1. S. M. Sze and M.K. Lee, **Semiconductor devices- Physics and Technology**, 3rd Edition, John Wiley & Sons, 2012.
2. Muller and Kammin, **Device Electronics for Integrated circuits**.
3. Vagica Vasileska and Stephen M. Goodnick, **Computational Electronics: Semiclassical and Quantum Device Modeling and Simulation**.
4. Shundri Oda & David Ferry, **Silicon Nanoelectronics**, CRC Press.

### References

1. S. M. Sze and Kwok K. Ng, **Physics of Semiconductor Devices**, 3rd Edition, John Wiley & Sons, 2002.
2. Ben G. Steetman and Sanjay Banerjee, **Solid State Electronic Devices**, 6th Edition, Prentice Hall, 2005
3. Robert F. Pierret, **Semiconductor Device Fundamentals**, Addison-Wesley Publishing, 1996
4. Donald A. Neamen, **Semiconductor Physics and Devices**, 3rd Edition, McGrawHill, 2003
5. Jasprit Singh, **Semiconductor Devices - Basic Principles**, John Wiley and Sons Inc., 2001